

Department of Mathematics, IIT Madras
Linear Algebra for Engineers (MA2031)
Assignment-2 (Linear Transformations)

Notations

- $R(T) = Im(T)$ denotes the image of a linear transformation T .
- The kernel of a linear transformation T is denoted by $N(T) = Ker(T)$.
- $null(T) = n(T)$ denotes the nullity of a linear map T , and $rank(T) = r(T)$ denotes the rank of a linear map T .
- Given $v \in V$, let $[v]_B$ denote the coordinate vector of v with respect to the basis B , and $[T(v)]_E$ denote the coordinate vector of $T(v)$ with respect to the basis E .
- $[T]_{B_2, B_1} = M_{B_1, B_2}(T)$ denotes the matrix representation of a linear map T with respect to the bases B_1 and B_2 .

1. Why is the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given below not a linear transformation?
 (a) $T(a, b) = (1, b)$ (b) $T(a, b) = (a, a^2)$ (c) $T(a, b) = (\sin a, 0)$
Ans: Look at (a) $T(0, 0)$ (b) $T(2, 2), 2T(1, 1)$ (c) $T(\pi/2, 0), 2T(\pi/4, 0)$
2. What are the all possible linear operators on $\mathbb{R}$? *Ans:* $T(x) = \alpha x$ for some α .
3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation with $T(1, 0) = (1, 4)$ and $T(1, 1) = (2, 5)$. What is $T(2, 3)$? Is T one-one? *Ans:* $(5, 11)$, Yes.
4. Can you construct a linear operator on $\mathbb{R}^2$ that maps the square with corners at $(-1, -1)$, $(1, -1)$, $(1, 1)$ and $(-1, 1)$ onto the square with corners at $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$?
Ans: No. Look at images of $(1, 1), (-1, -1)$.
5. In each of the following, construct a linear transformation T if it exists.
 - (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$; $T(1, 1) = (1, -1)$, $T(0, 1) = (-1, 1)$, $T(2, -1) = (1, 0)$. *Ans:* No. $T(2, -1) = ?$
 - (b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $T(1, 0, 3) = (1, 1)$, $T(-2, 0, -6) = (2, 1)$.
Ans: No. $T(-2, 0, -6) = ?$
 - (c) $T : \mathbb{R}_3[t] \rightarrow \mathbb{R}$; $T(a + bt^2) = 0$ for any $a, b \in \mathbb{R}$.
Ans: $T(a + bt + ct^2 + dt^3) = b + c$ and others.
 - (d) $T : C^1[0, 1] \rightarrow \mathbb{R}^2$; $T(u) = (\int_0^1 u(t) dt, u'(0))$. *Ans:* This T .
6. Determine $rank(T)$ and $null(T)$ in each of the following cases:
 - (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$; $T(a, b) = (a - b, 2b)$.
 - (b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$; $T(a, b) = (a + b, a - b, 0)$.
 - (c) $T : \mathbb{R}_2[t] \rightarrow \mathbb{R}_3[t]$; $T(p(t)) = t p(t) + p'(t)$.*Ans:* $rank(T), null(T) =$ (a) 2, 0 (b) 2, 0 (c) 3, 0.
7. Let V be the vector space of all functions from $\mathbb{R}$ to $\mathbb{R}$, having derivatives of all orders. Let $T : V \rightarrow V$ be the differential operator: $Tx = x'$. What is $N(T)$?
Ans: Space of all const. functions.
8. Let V and W be finite dimensional vector spaces. Let $T : V \rightarrow W$ be a linear transformation. Give reasons for the following:
 - (a) $rank(T) \leq \min\{\dim(V), \dim(W)\}$.

- (b) T is onto implies $\dim(W) \leq \dim(V)$.
- (c) T is one-one implies $\dim(V) \leq \dim(W)$.
- (d) $\dim(V) > \dim(W)$ implies T is not one-one.
- (e) $\dim(V) < \dim(W)$ implies T is not onto.

9. Give an example for each of the following:

- (a) A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $N(T) = R(T)$.
- (b) Linear transformations $S \neq T$ with $N(S) = N(T)$ and $R(S) = R(T)$.

Ans: (a) $T(a, b) = (a - b, a - b)$ (b) $S(a, b) = (a, b)$, $T(a, b) = (b, a)$.

10. Is $V := \{A \in \mathbb{F}^{n \times n} : \text{tr}(A) = 0\}$ a vector space? Ans: Yes.

11. Show that the map $T : \mathbb{F}^{n \times n} \rightarrow \mathbb{F}$ defined by $T(A) = \text{tr}(A)$ is a linear map. Show that T is an onto map. Also, find $\text{null}(T)$. Ans: $\text{null}(T) = \dim_{\mathbb{F}}(V) = n^2 - 1$.

12. Construct a matrix $A \in \mathbb{R}^{2 \times 2}$ with $\text{tr}(A^2) < 0$. Ans: $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

13. Let $\{v_1, \dots, v_n\}$ be an ordered basis of a vector space V . Prove that the linear transformation $T : V \rightarrow \mathbb{F}^n$ given by $T(\alpha_1 v_1 + \dots + \alpha_n v_n) = (\alpha_1, \dots, \alpha_n)$ is an isomorphism.

14. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(a, b) = (a - b, a, 2a - b)$. Let B be the standard basis of $\mathbb{R}^2$. Take $C = \{(1, 2), (2, 3)\}$ and $D = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$ as ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Compute $[T]_{D,B}$ and $[T]_{D,C}$.

Ans: $[T]_{D,B} = \begin{bmatrix} -1/3 & 1/3 \\ 0 & 1 \\ 2/3 & -2/3 \end{bmatrix}$; $[T]_{D,C} = \begin{bmatrix} 1/3 & 1/3 \\ 2 & 3 \\ -2/3 & -2/3 \end{bmatrix}$.

15. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(a, b, c) = (b + c, c + a, a + b)$. Determine $[T]_{E,B}$ where

- (a) $B = \{(1, 0, 0), (0, 0, 1), (0, 1, 0)\}$, $E = \{(0, 0, 1), (1, 0, 0), (0, 1, 0)\}$.
- (b) $B = \{(1, 1, -1), (-1, 1, 1), (1, -1, 1)\}$, $E = \{(-1, 1, 1), (1, -1, 1), (1, 1, -1)\}$.

Ans: (a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

16. Let $T : \mathbb{F}_2[t] \rightarrow \mathbb{F}_3[t]$ be given by $T(p(t)) = t p(t)$. Consider the ordered bases $B = \{1 + t, 1 - t, t^2\}$ for $\mathbb{F}_2[t]$ and $E = \{1, 1 + t, 1 + t + t^2, t^3\}$ for $\mathbb{F}_3[t]$.

Find the matrix $[T]_{E,B}$.

Ans: $\begin{bmatrix} -1 & -1 & 0 \\ 0 & 2 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

17. Construct matrices $A, B \in \mathbb{F}^{n \times n}$ so that $\text{tr}(AB) \neq \text{tr}(A)\text{tr}(B)$. Ans: $A = I = B$.

18. For $\mathbb{R}^{3 \times 1}$, consider the ordered bases $O = \{[1, 0, 1]^t, [1, 1, 0]^t, [0, 1, 1]^t\}$ and $N = \{[1, -1, 1]^t, [1, 1, -1]^t, [-1, 1, 1]^t\}$. Let T be the linear operator on $\mathbb{R}^{3 \times 1}$ whose matrix

representation with respect to the standard basis of $\mathbb{R}^{3 \times 1}$ is given by the matrix $\begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Find the matrix $[T]_{N,O}$ and verify that $\left[T \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right]_N = [T]_{N,O} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_O$.

Ans: $[T]_{N,O} = \frac{1}{2} \begin{bmatrix} 2 & 3 & 3 \\ 2 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$.